IE406: Machine Learning Lab Assignment-4
Name: Parth Karena
ID: 202301472

Lab-5: Softmax-Based Multi-Class Logistic Regression

Task: Softmax-Based Multi-Class Classification and implement with SGD

- The Training and Validation Loss for each epoch is shown

Statistics :-

Questions :-

1. Which model performed better?

Softmax Logistic Regression generally performs better for multi-class classification because it directly models the probability distribution over all classes simultaneously.

2. Is Softmax more stable than One-vs-Rest?

Yes, Softmax is more stable because it ensures that the predicted probabilities across all classes sum to 1. In contrast, One-vs-Rest trains separate classifiers independently, which can lead to inconsistent probability estimates.

3. Which approach required fewer classifiers?

Softmax logistic regression requires **only one model**, while One-vs-Rest requires **one classifier per class**. For KKK classes, One-vs-Rest requires **K classifiers**.

Conclusion :-

In this lab, Softmax-based multi-class logistic regression was implemented from scratch using NumPy and trained using Stochastic Gradient Descent. The model was evaluated using multiple classification metrics including accuracy, balanced accuracy, AUROC, and confusion matrix.

The Softmax approach provides a unified framework for multi-class classification and generally produces more consistent probability estimates compared to the One-vs-Rest approach.